

BOSHEALTH: EMERGENCY RESPONSE APP SYSTEM



Saving Time, Saving Lives

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AGENDA

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INTRODUCTION

“You can’t prevent emergencies, but you can design systems that are prepared for them.”

The BOSHealth App is a key component of the Smart Emergency Response System, designed to reimagine emergency healthcare delivery. Our project includes multiple interfaces that work together to reduce delays, improve communication, and optimize resource allocation during medical emergencies.

Project Deliverables:

- **Live Dashboard:** Panel displaying real-time ambulance and patient locations, severity of cases, and hospital capacity.
- **Executive Dashboard:** Analytics view of trends in ETA based on weather/traffic, patient severity, and high incident zones.
- **BOSHealth App (3 Interfaces):**
 - **Patient Interface:** A mobile app for individuals to request ambulances, track ETA and share relevant medical information.
 - **Dispatcher Interface:** A system used by ambulance drivers to accept calls, navigate routes, and sync vitals with hospitals.
 - **Hospital Interface:** A real-time interface for ER team to track new incoming patients, view their medical history and live vitals, and decide who needs urgent care first.

DASHBOARDS OVERVIEW

As the basis of the BOSHealth system, two dashboards have been designed using existing emergency response data. These dashboards offer live ops view and executive insights, providing a clear picture of how emergency services are currently performing, and where improvements are needed.

Live Dashboard: A real-time operational view built for users, EMS teams and hospital staff.

- Live ambulance location tracking
- Severity-based categorization
- Hospital bed availability and status
- ETA and dispatch status monitoring

Executive Dashboard: A high-level strategic view for administrators, providing:

- Trends in dispatch times and hospital loads
- Heatmaps of high-incident zones
- Avg. monthly trips to balance loads
- ETA trends varying by weather/traffic conditions

These dashboards lay the foundation for BOSHealth system, offering **data-driven insights** on how emergency care works today to better prepare us to build smarter and connected care experience in future.

REAL-WORLD SCENARIO: WHY BOSHEALTH MATTERS!

In January 2024, a tragic incident in **Winthrop, Massachusetts**, revealed the consequences of emergency response delays. A **2-year-old girl** passed away after an ambulance failed to arrive in time.

At least **23 minutes** elapsed between the initial **911 call** and the child's arrival at **Massachusetts General Hospital**. The closest available ambulance was reportedly **20 minutes away**. To save her, a police officer transported the child in his own SUV, while two EMTs administered care during the drive. Despite their efforts, the child could not be saved, a devastating outcome that highlights a fundamental issue (NBC Boston, 2024).

This case highlights **breakdowns** in emergency response:

- **Lack of ambulance proximity**
- **Poor communication** between responders and hospitals

BOSHealth is built to close these gaps with technology that empowers faster decisions, smarter routing, and better hospital preparedness, because in emergencies, every second matters.

USER PERSONA/ TARGET USERS

Persona 1:

Name: Niya Sharma

Age: 60

Medical condition: Hypertensive

Tech comfort: Moderate

Family: Lives alone, daughter lives 30 minutes away

Needs:

- Get help quickly without calling 911 manually
- Track ambulance location and arrival time
- Automatically notify daughter

Pain points:

- To repeat medical history to new responders
- Delays in ambulance arrival
- No real time tracking

BOSHealth is primarily designed for a growing community of independent and tech aware **older adults** living with various chronic health conditions. Most of them live alone or might have limited family support nearby, they rely mostly on these digital tools to manage their care and to feel better prepared for emergencies.

Persona 2:

Name: Ryan P.

Age: 26

Medical condition: Mild asthma

Tech comfort: High

Living Situation: Lives with friends in shared apartment

Needs:

- Instantly share medical information
- Stay connected with family while receiving care.
- Track live status of ambulance.

Pain points:

- Lack of immediate coordination between ambulance and hospital
- Anxiety about uncertainties during an emergency
- Delay in sharing medical history

BOSHealth also serves **young, tech savvy generation** including university students, professionals and parents, who value clear and instant updates during emergencies. Some

of the features like live tracking, estimated ETA updates, vital sync with hospitals can provide them with better comfort for themselves and for their loved ones. Together, these groups represent the target audience for our app. By understanding their needs better, we have designed an app that would deliver urgent and reliable care during emergency.

APP INTERFACE

KEY FEATURES & FUNCTIONALITIES

User Registration

- Sign-up via email, Google, or Apple
- Enter personal details (DOB, phone, email, address)
- Add medical history: conditions, allergies, medications
- Provide insurance and emergency contact info.

Location Permissions

- App request's location permission (precise or while using)
- Live location is used to identify nearby ambulances and hospitals

Ambulance Request

- Choose between Basic Life Support (BLS) or Advanced Life Support (ALS)
- View nearest ambulance options with distance and ETA
- Initiate a 911 emergency request directly through the app

Live Tracking & ETA

- Track ambulance in real time
- Get estimated time of arrival (e.g., "Help is on the way! ETA is 5 min")
- Display of selected hospital and its distance

Emergency Contact Notification

- Automatic notification to listed emergency contact
- Location and status shared with them

Vitals & Medical Info Sharing

- Vital signs are continuously monitored and transmitted to the receiving hospital
- Paramedics and ER team can access critical medical history in advance

Post-Pickup Journey

- Patient can see ongoing updates about their transport
- Emergency contact can also monitor progress

Access to a detailed app write up and design artifacts

- [Full Write Up](#)
- [Patient App - Prototype](#)
- [Hospital App - Prototype](#)
- [Dispatcher App Prototype](#)

USER JOURNEY FLOW

Workflow Scenario

This section illustrates a hypothetical patient journey based on the proposed BOSHealth app design. While the app is currently in the initial design stage, this user workflow demonstrates how its features could work together in an actual emergency.

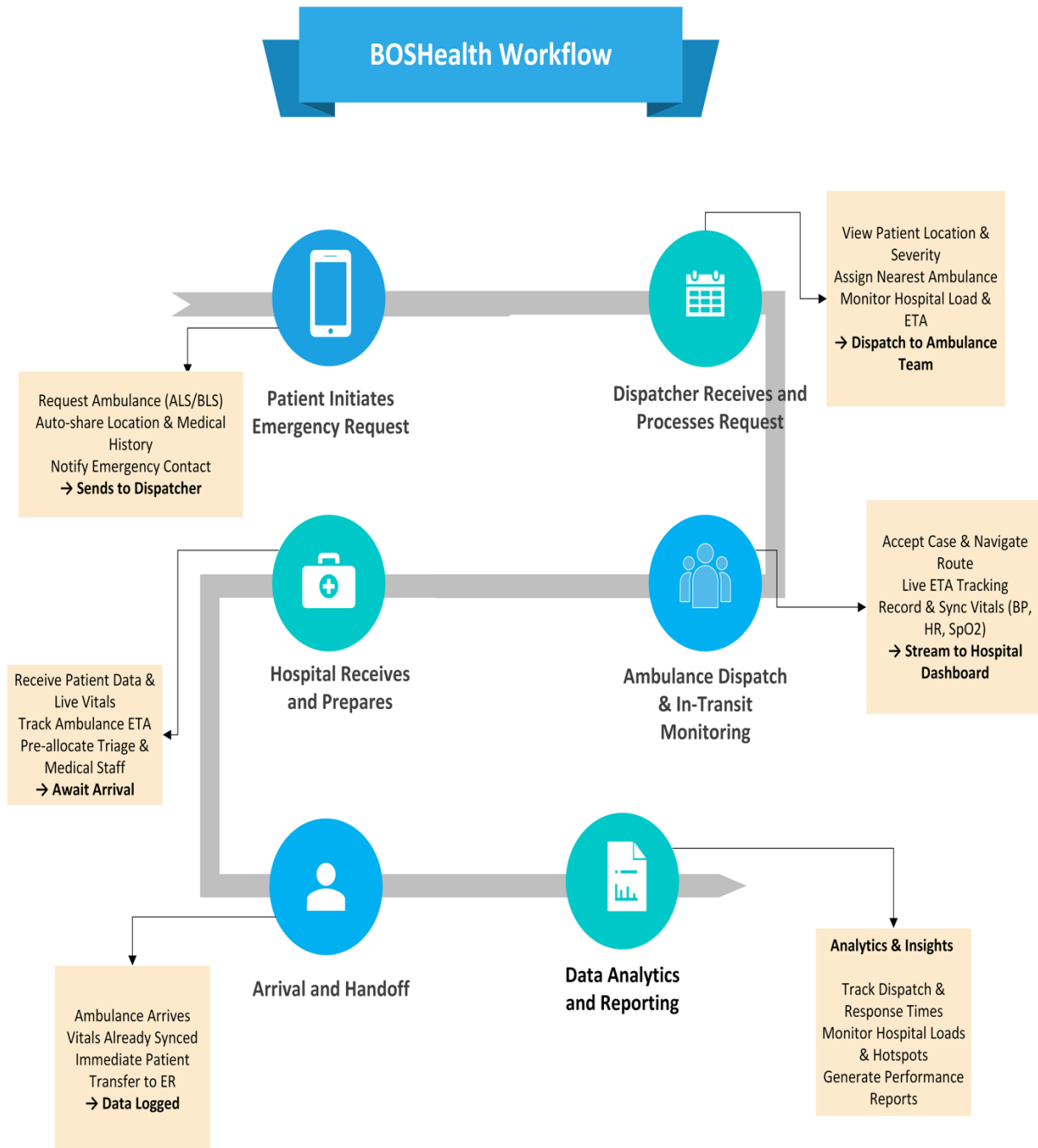
Profile:

- **Name:** Mark Scout
- **Location:** Boston, MA
- **Medical Condition:** Type 2 diabetes (Medication- Glipizide)

Workflow:

- **Sign-Up:** Mark creates an account, enters personal and medical details and adds his emergency contact (Helly R).
- **Enable Location Access:** He enables precise location tracking, allowing BOSHealth to suggest him with the nearest ambulance.
- **Request Help:** Mark experiences dizziness and uses the app to request an ambulance. Based on his condition, he selects BLS (Basic Life Support).
- **Ambulance Dispatched:** A nearby team accepted the request. The app shows a live ETA of 5 minutes.
- **Vitals Monitored and synced:** During transport, paramedics record and sync vitals (HR, BP, SpO2) with the hospital dashboard.
- **Pt. Reaches Hospital:** The ER receives live updates and medical history in advance. The ER is prepared beforehand for his arrival.

Workflow diagram



LESSONS LEARNED

Through the development and integration of the BOSHealth Emergency Response App, along with its related dashboards, the team learned a number of practical and technical lessons:

- **User-Centric Design is Critical:** In an emergency, where people are under stress, and they don't have time to learn or navigate complicated screens. We learned that interfaces must be simple, intuitive, and easy to operate under stress. Small changes can make a huge difference in usability like, large buttons, obvious alerts, and reducing distracting design elements.
- **User Feedback:** Discussions with potential users transformed how we initially approached the app's emergency request design. Their insights pinpointed practical obstacles that we had not previously considered, like sending alerts to family members.
- **Real-World Network Challenges:** We also anticipated some troubling issues with unreliable network coverage, particularly in the ambulance transport environment. While we are not currently building any offline support, we had to remember emergency systems need to take connectivity into account due to variations in communication, and we flagged this for future development.
- **Prioritizing Features for Impact:** During our initial brainstorming, we were excited to potentially design an app with awesome features like AI and predictive outputs or uploading all previous patient history. However, our feedback and discussions enabled us to understand the need for focus on must-haves, like live tracking of the ambulance, estimated time of arrival (ETA) updates, syncing vitals, and emergency contact alerts, the items users will need the most during emergencies.
- **Connecting Interfaces and data sync:** It was a challenge to design the connection and interaction between ambulance, users and hospitals.

FUTURE RECOMMENDATIONS

To further enhance and develop BOSHealth, we recommend:

- **AI-Driven Recommendations:** Leverage AI models to better predict ambulance arrival times with better accuracy, by incorporating real time traffic, weather, and historical dispatch data under similar conditions. This could improve routing optimization and hospital selection during critical emergencies.
- **Multilingual & Accessibility Features:** Allow for multilanguage interfaces, voice interactions, and the necessary design features for the application to be used by larger range of users with disabilities or language barriers.
- **Rural and Low-Connectivity Access:** Customize the system for use in rural areas with limited and/or intermittent connectivity.
- **Integration with EHR Systems:** Provide BOSHealth the ability to connect to hospital Electronic Health Records (EHR) systems to allow for seamless, secure, and immediate transfer of patient information at time of arrival.

CONCLUSION

"Delayed response isn't a glitch, it's a gap we're here to fix."

The BOSHealth Emergency Response App is a disruptive innovation in emergency care! By bringing together patients, EMTs, and hospitals through real-time data and easy-to-use interfaces, BOSHealth addresses one of the major bottlenecks in health care, which is time, and the need for more timely response.

BOSHealth saves the time wasted when dispatchers must "guess" what actions to take and get emergency rooms prepared before the patient arrives. It is a human-centric experience system that saves lives under critical time constraints.

BOSHealth creates a new benchmark for connected emergency response options that can be applied in smart cities such as Boston and creates a pathway for scalable life-saving systems across the world.

Note:

We have included a presentation with this document for a visual walkthrough of our project.

